

Jacob Berger

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EDUCATION

Oregon State University Honor's College

Bachelor of Science in Applied Computer Science

Bachelor of Science in Computational Mathematics

• **Concentration:** Artificial Intelligence | **GPA:** 3.9+ | Accelerated Master's Program

• **Relevant Coursework:** Machine Learning & Data Mining, Intro to Artificial Intelligence, Analysis of Algorithms, Mathematics of Data Science (Graduate), Numerical Analysis, Linear Algebra II

Corvallis, OR

September 2022 – June 2026

RESEARCH

Honors Thesis: Dynamic Constraints for LLMs

Researcher (Advisor: Dr. Lizhong Chen, STAR Lab)

• Leading an honors thesis project to develop a dynamic constraint system for large language models, improving text generation for applications in computational semantics.

Corvallis, OR

June 2025 – Present

Senior Capstone Project: AI for Home Automation

Project Team Member

• Collaborating in a senior design team to integrate Large Language Models with the Home Assistant IoT platform, enabling users to control devices and create automations using natural language.

Corvallis, OR

September 2025 – Present

EXPERIENCE

AI Startup (OSU STAR Researchers)

Language Model and Software Engineer

• Developed a full data pipeline for training and deploying specialized large language models.

• Led and co-authored the team's approved Phase I Small Business Innovation Research (SBIR) proposal.

• Engineered and deployed cloud infrastructure on AWS, GCP, and Vercel to train and host services.

Corvallis, OR

March 2023 – Present

MU Lanes and Games

Head Mechanic

• Managed maintenance projects and diagnostics for complex electromechanical systems.

Corvallis, OR

December 2022 – Present

MU Student Activities and Advisory Board

Vice-Chairman

• Oversaw budget preparation and facilitated strategic board meetings for OSU's Memorial Union.

Corvallis, OR

June 2024 – Present

SKILLS

• **Languages:** Python, C++, Java, Javascript, Matlab, x86 Assembly

• **Machine Learning & AI:** PyTorch, Open-source LLMs, LLM Fine-tuning, Controlled Text Generation

• **Software & Tools:** Linux, Git, Amazon Web Services (AWS), Google Cloud Platform (GCP)

• **Languages (Human):** Fluent in Spanish (9 years of language immersion education)

AWARDS

• **Putnam Math Competition Award:** Top 3 Finisher at Oregon State University.

• **Finley Academic Excellence Scholarship:** \$20,000 scholarship recipient for academic merit at OSU.

EXTRACURRICULARS

• **Competitive Math:** Competitor in Oregon State Putnam and COMAP Math Modeling competitions.

• **Competitive Sports:** Competitive Sports: Founded OSU's national collegiate bowling team (All-Star qualifier); served as Rose City Rowing Club team leader (2-time national qualifier).